

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: USB3 Gen T Link Command Aggregation and Coalescing updates

Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

- Defining the expected coalescing rules when Link Command Aggregation is disabled
- Clarifying the changes for Gen T with regards to the link command index expectations, which updates the Gen X requirement for expecting them to be consecutive for each counter
- Redefining *tLC_AggrThreshold* range with minimum and maximum limits. The range is defined for allowing Link Commands aggregation for 4x 1KB Data Packets time intervals in Gen4 Symmetric Link or Gen3 Dual Link to
 - Minimum for 4x 1KB packets without USB4 Credit Grant packets interleaving. [supporting back to back unidirectional traffic \(IN or OUT\)](#).
 - Maximum for 4x 1KB packets with USB4 Credit Grant packets interleaving for every packet (IN+OUT).

Benefits as a result of the changes:

1. Clarifications would assist sorting the error cases, and avoiding related future inter-operability issues.
2. Updating *tLC_AggrThreshold* range to enable performance optimization with USB3 Gen T.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:

NA

An analysis of the hardware implications:

NA

An analysis of the software implications:

NA

An analysis of the compliance testing implications:

NA

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Actual Change

(a). 9.4.1.5 Link Command Aggregation and Coalescing

From Text:

An Internal USB3 Gen T Component shall support Link Command Aggregation and Coalescing. Once the link is in LTSSM.U0, the Internal USB3 Gen T Component shall turn on Link Command Aggregation if the *Link Commands Aggregation Enable* bit is set to 1b in the USB3 Gen T Adapter Configuration Space.

An Internal USB3 Gen T Component shall send a Link Command (if it has one to send) without waiting for *tLC_AggrThreshold* upon any of the following:

- Entry into LTSSM.U0.
- After a *tLC_AggrThreshold* time interval passes without any Link Commands.
- If the Link Command is not an LGOOD_n, LCRD1_x, or LCRD2_y.

All subsequent Link Commands shall be sent at *tLC_AggrThreshold* intervals except as listed above. The Internal USB3 Gen T Component shall send the last LGOOD_n/LCRD1_x/LCRD2_y Link Commands as per the Header Packets received during the *tLC_AggrThreshold* period.

To Text:

An Internal USB3 Gen T Component shall support Link Command Aggregation and Coalescing. Once the link is in LTSSM.U0, the Internal USB3 Gen T Component shall turn on Link Command Aggregation if the *Link Commands Aggregation Enable* bit is set to 1b in the USB3 Gen T Adapter Configuration Space.

Note: Link Commands Aggregation permits sending LCRD_1_x, LCRD2_y, and LGOOD_n with the most recent index (i.e. skipping over indexes) regardless of the value of the Link Commands Aggregation Enable bit. The value of tLC_AggrThreshold is defined for keeping the Link Command maximal rate as one for every 4KB of transferred user payload. While the Link Commands Aggregation Enable bit is set to 0b, the rules below for Link Command Aggregation and Coalescing still apply, except that the Internal USB3 Gen T Component does not need to maintain tLC_AggrThreshold intervals between Coalesced Link Command Packets.

An Internal USB3 Gen T Component shall send a Link Command (if it has one to send) without waiting for *tLC_AggrThreshold* upon any of the following:

- Entry into LTSSM.U0.
- After a *tLC_AggrThreshold* time interval passes without any Link Commands.
- If the Link Command is not an LGOOD_n, LCRD1_x, or LCRD2_y.
- *If the Link Commands Aggregation Enable bit is cleared.*

All subsequent Link Commands shall be sent (if such are needed) at *tLC_AggrThreshold* intervals except as listed above while the USB3 Gen T LTSSM is in U0. *The transmission of a Link Command may be delayed in the following cases:*

- *The USB3 Gen T Port is in the process of transmitting a USB3 Data Packet*
- *The USB4 Path associated with the USB3 Gen T Port is blocked by insufficient USB4 credits*

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- The USB4 Path associated with the USB3 Gen T Port is blocked by USB4 packets being transmitted on a different USB4 path on the USB4 link.

The Internal USB3 Gen T Component shall send the last LGOOD_n/LCRD1_x/LCRD2_y Link Commands as per the Header Packets received during the *tLC_AggrThreshold* period, then update the Rx Header Sequence Number and Rx Buffer Credit indexes accordingly.



IMPLEMENTATION NOTE

A Gen T Port needs to avoid transmitting LCRD1_x or LCRD2_y after freeing a packet buffer before an LGOOD_n is transmitted for the corresponding packet to which the LCRD1_x/LCRD2_y is associated.

(b). 9.4.1.7 USB3 Gen T Link Timing Parameters

From Text:

Name	Description	Min	Max	Units
tLC_AggrThreshold	Minimum time before any pending Link Commands shall be sent	Gen 4: 500 Gen 3: 1000	--	ns

To Text:

Name	Description	Min	Max	Units
tLC_AggrThreshold	Time before any pending Link Commands shall be sent	Gen 4: 458 440 Gen 3: 914 880	Gen 4: 480 950 Gen 3: 1080 2100	ns